

Logistic Regression basic Work flow :

classmate

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A classification problem

1. Input features

$$X = [x_1, x_2, \dots, x_n]$$

n = no of features

2. Linear Combination (z-score)

→

$$Z = W^T X + \beta$$

Weights

Bias

4. Threshold

(usually 0.5)

$p \geq 0.5 \Rightarrow$ predict 1

$p < 0.5 \Rightarrow$ predict 0

3. Sigmoid Function

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

- Converts value of z into between 0 to 1

Value of z : $-\infty$ -10 -5 0 5 10 $+\infty$

Value of p : < closer to 0 > 0.5 < closer to 1 >

5. Prediction

$$\hat{y} = 1 \text{ or } 0$$

★ How the model is trained (Find Best W and β)

1. Likelihood (How well β and W do for prediction)

our job is to maximize Likelihood

(just like we minimize Loss func in Linear reg (GD))

For its data point

$$P(y^i | x^i) = p_i^{y_i} (1 - p_i)^{1 - y_i}$$

While predicting for class 1

this value will remain 1 and

vice versa for class 0

$$L(W, \beta) = \prod_{i=1}^n p_i^{y_i} (1-p_i)^{1-y_i}$$

this Likelihood which we have to maximize
(we multiply p as they come independent events)
each record is independent

2. Log-Likelihood (Take log for easy computation, it does not affect the result)

log converts multiplication into addition

$$\log L(W, \beta) = \sum_{i=1}^n [y_i \log(p_i) + (1-y_i) \log(1-p_i)]$$

Maximizing $L \Leftrightarrow$ Maximizing $\log L$

3 Negative Log-Likelihood / Log Loss IMP *

$$NLL(W, \beta) = -\log L(W, \beta)$$

GD is an optimization algo which can only minimize, that's why we flip the sign.

we flip the sign as GD normally moves to minimize
After flipping we are moving in opposite direction!

So maximize Log-Likelihood = minimize (-Log Likelihood)

^{also} This Log Loss is called Binary Cross Entropy (part of PGD which can only minimize)

4 Optimization (find best w and β)

Repeat

Choose $W, \beta \rightarrow$ Compute p 's (using sigmoid) $\rightarrow$ Compute Log Loss

Update
 W, β

Compute gradients
(how much to change W, β)

5 Final model & Prediction

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

~~X_n~~

$X_{\text{new}} \rightarrow z = W^T \cdot X_{\text{new}} + \beta \rightarrow$ Sigmoid $\rightarrow$ Threshold $\rightarrow$ Predict
(get p) 0.5

Formulas

$$P(Y|X) = \frac{e^{CWX+\beta}}{1 + e^{CWX+\beta}}$$

Vector form

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_n \end{bmatrix} \quad W = \begin{bmatrix} w_1 \\ w_2 \\ w_n \end{bmatrix} \Rightarrow W^T \cdot X = \begin{bmatrix} w_1 & w_2 & w_n \end{bmatrix} \begin{bmatrix} x_1 x_2 x_n \end{bmatrix}$$